1. Problem

Earthquakes are unpredictable natural disasters that cause severe damage to life and infrastructure. Traditional prediction methods are not accurate enough due to the complexity of seismic patterns and lack of real-time analysis.

There is a need for a smart system that can:

- Analyze large seismic datasets
- Identify hidden patterns
- Predict earthquake occurrence and magnitude
- Help in early warning and disaster preparedness

2. Project Name

ImpactSense - Earthquake Impact Prediction

3. Goal

The main goal of this project is to develop a machine learning-based system that:

- Predicts earthquake magnitude or occurrence
- Improves prediction accuracy using advanced algorithms
- Provides insights for disaster management
- Helps reduce the impact of earthquakes through early analysis

4. Proposed Solution

ImpactSense is an AI-based system that uses historical earthquake data to predict future seismic activity. The system applies machine learning algorithms like **Random Forest** and **Gradient Boosting** to analyze features such as location, depth, and magnitude.

The model is trained on past data and can:

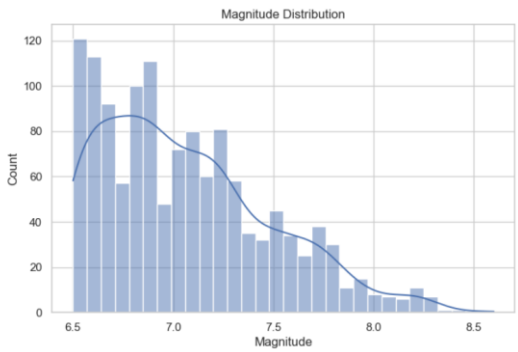
- Predict earthquake intensity
- Identify high-risk zones
- Provide analytical insights through visualizations

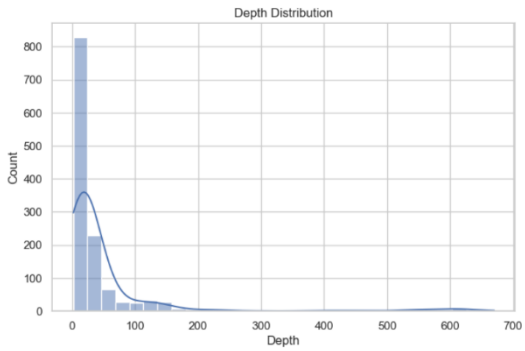
A web-based interface is implemented to allow users to input data and receive real-time predictions in an interactive and user-friendly manner.

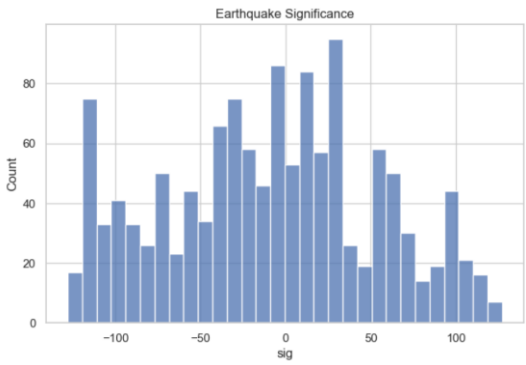
5. Implementation (Weekly Tasks)

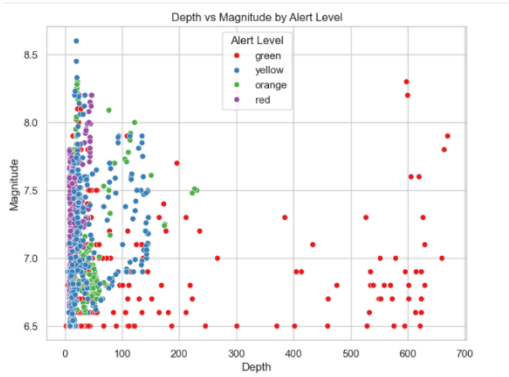
5.1 Week 1 – Data Collection & Understanding

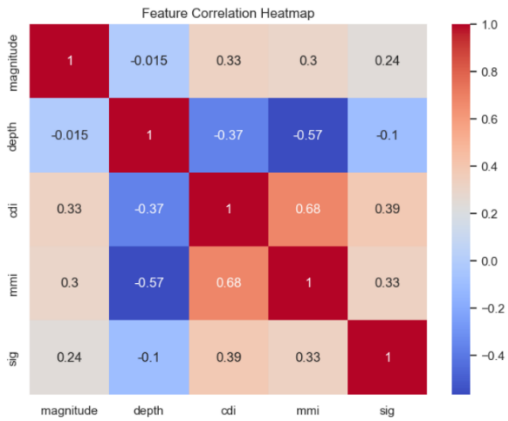
- Collected earthquake dataset from sources like USGS/Kaggle.
- Understood dataset structure (features like magnitude, depth, cdi, mmi, sig and alert)
- Performed initial data exploration

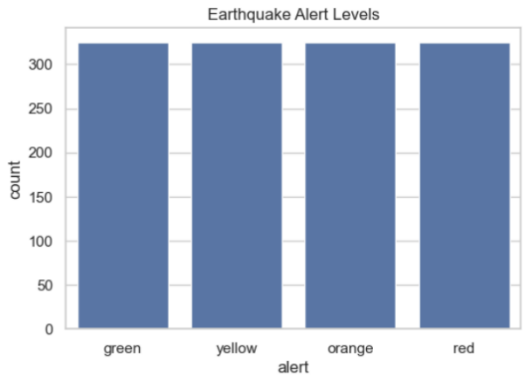










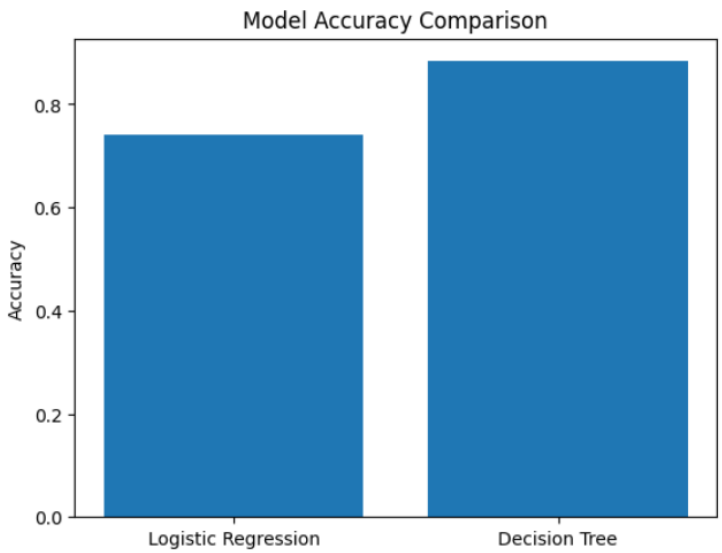


5.2 Week 2 – Data Preprocessing

- Handled missing values
- Removed irrelevant columns
- Normalized and scaled data
- Performed data cleaning

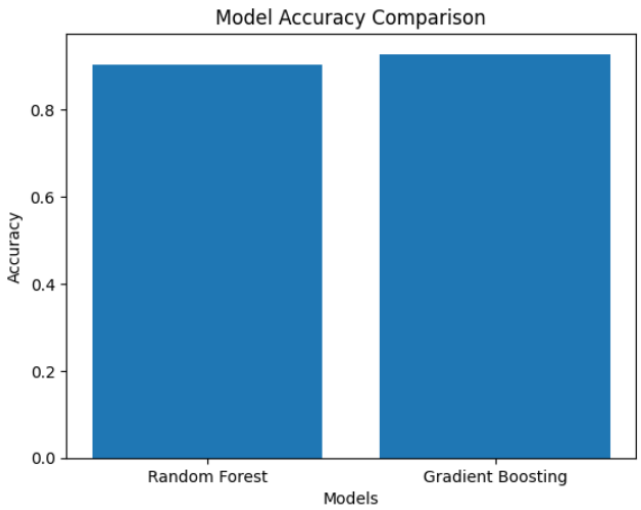
5.3 Week 3 – Baseline Model Training

- Logistic Regression, Decision Tree
- Basic accuracy/MAE analysis.



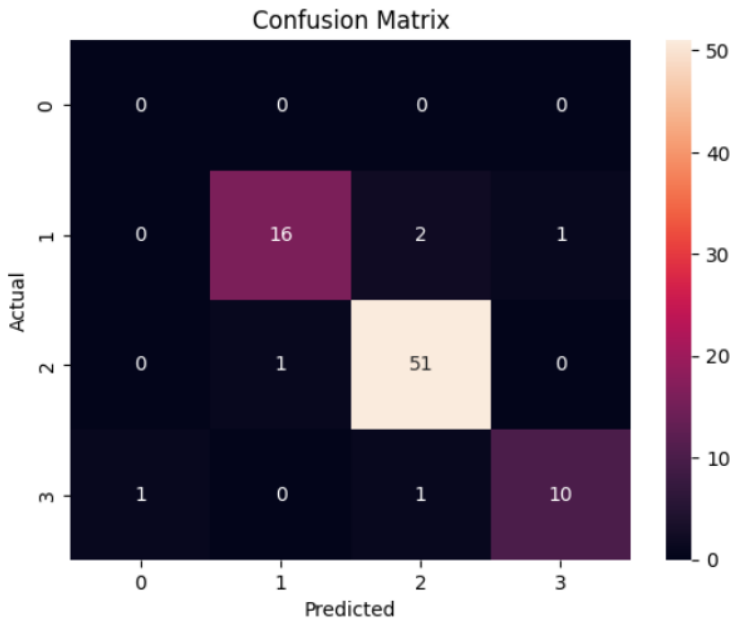
5.4 Week 4 – Advanced Model Training

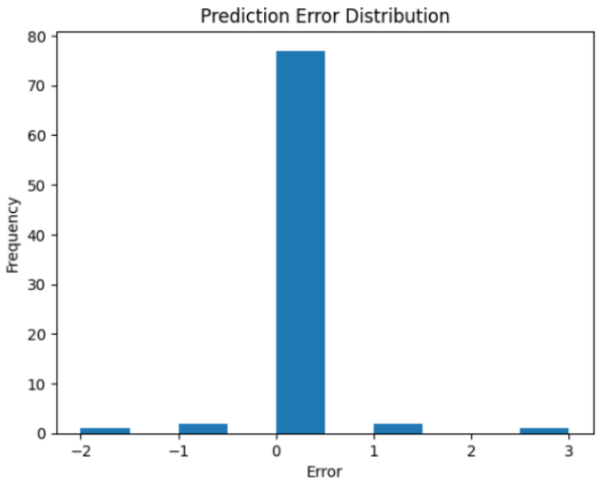
- Random Forest, Gradient Boosting
- Cross-validation, hyperparameter tuning.

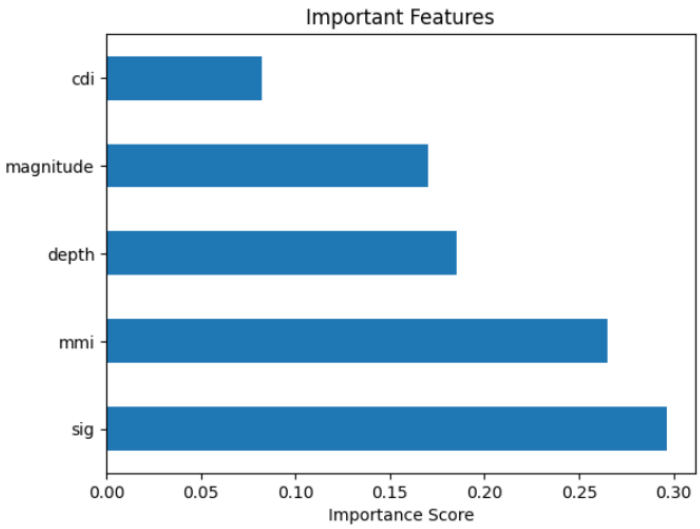


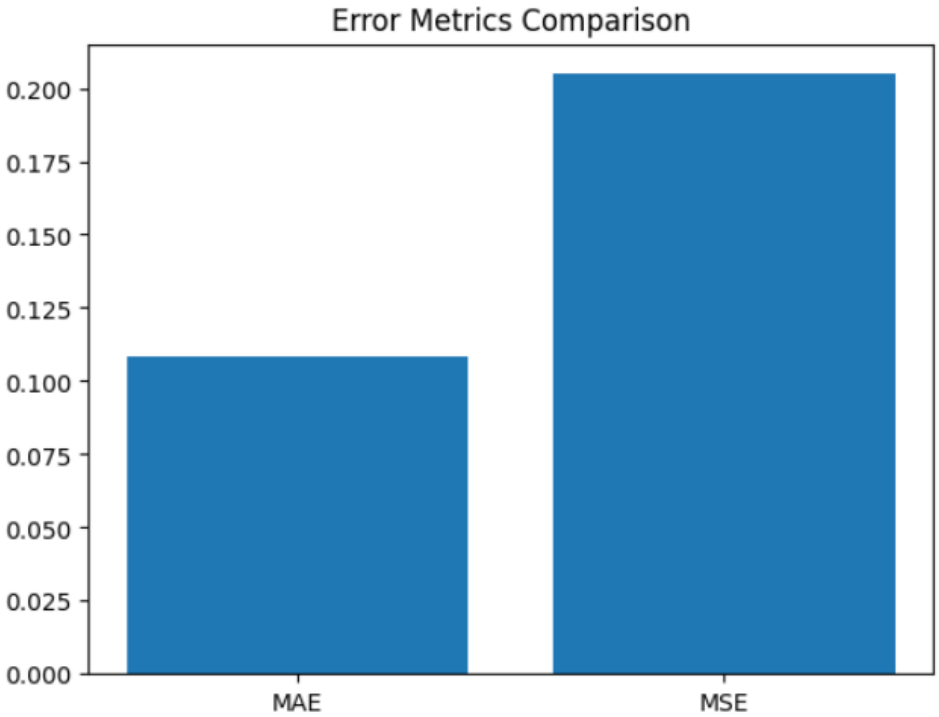
5.5 Week 5 – Evaluation & Explainablility

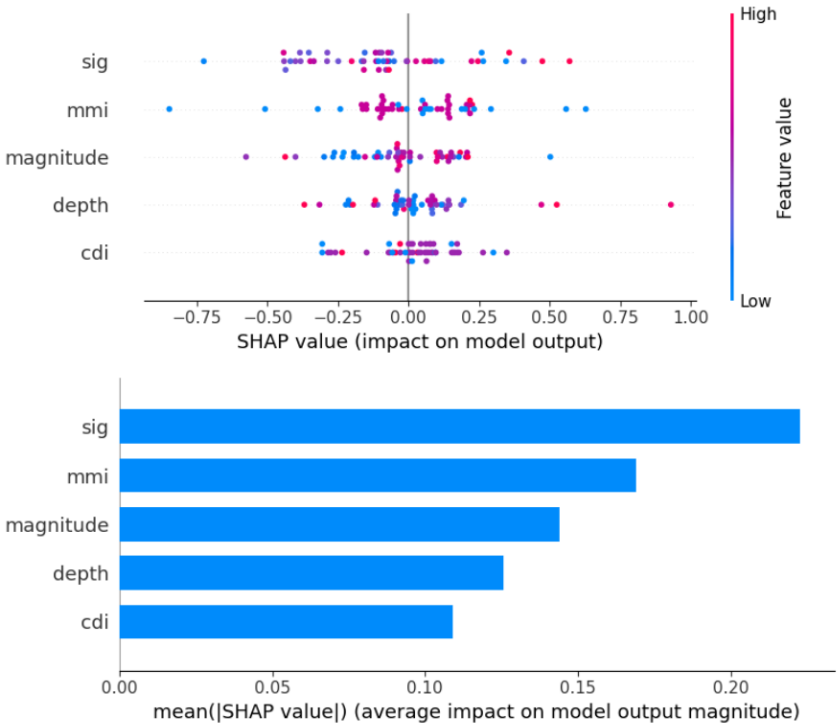
- Generate confusion matrix, MSE/MAE plots.
- Feature importance, SHAP values.
- Improved accuracy (up to ~95% using Gradient Boosting).








Error Metrics Comparison
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
MAE
MSE


sig
mmi
magnitude
depth
cdi
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
SHAP value (impact on model output)
High
Feature value
Low
sig
mmi
magnitude
depth
cdi
0.00
0.05
0.10
0.15
0.20
mean(|SHAP value|) (average impact on model output magnitude)

5.6 Week 6 – UI Development

- Built a simple web interface using Flask
- Connected model with frontend
- Enabled user input for predictions
- Displayed prediction results clearly.

5.6 Week 7 – Testing & Improvements

- Tested edge cases to ensure system reliability.
- Improved model performance & UI functionality.

5.6 Week 8 – Final Reports & Presentation

- Charts, visuals, final results, slides and PDF.

6. Tech Stack Used

Deployment Platforms :

- Frontend Hosting : Vercel
- Backend Hosting : Render

Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Python
- Flask

Machine Learning Models:

- Random Forest
- Gradient Boosting (Best performing)

Libraries:

- NumPy
- Pandas

- Matplotlib
- Seaborn
- Scikit-learn
- Folium
- SHAP

Tools:

- Jupyter Notebook
- VS Code

7. Future Scope

- Integration with real-time seismic data APIs
- Use of Deep Learning models for better accuracy
- Mobile application development
- Early warning alert system
- Cloud scaling using Docker and Kubernetes

8. Conclusion

The *ImpactSense – Earthquake Prediction Model* is a useful project that shows how Machine Learning can be applied to real-world problems like earthquake prediction. The system analyzes seismic data and provides predictions that can help in early awareness and better disaster preparation.

The project uses a modern approach by deploying the frontend on Vercel and the backend on Render, which makes the system fast, scalable, and easy to use. It also includes tools like SHAP to understand how the model is making predictions, which improves trust and transparency.

Although earthquake prediction is a difficult task and cannot be 100% accurate, this project provides a strong base for future improvements. With better data and advanced models, the system can become more accurate and reliable.

It represents a meaningful step toward building intelligent, scalable, and real-world AI solutions for disaster management.